

Week 4

NPTEL

[Powered Widgets](#)
[About the course](#)
[Announcements](#)
[My Bookmarks](#)
[Q&A](#)
[Manage Exam](#)

Introduction To Operating Systems

Course Progress 50.55% (3 of 7)

- ☒ Introduction
- ☐ Create Execute and Exit from Processes
- ☐ System Calls for Process Management
- ☐ Week 3: Lecture notes
- ☐ Week 3 Feedback Form : Introduction to Operating Systems!
- ☐ Quiz: Week 3 : Assignment 3
- ☐ Week 3: Assignment Solution
- Week 4: Interrupts**
 - ☐ Interrupts
 - ☐ Interrupt Handling
 - ☐ Software Interrupts and System calls
 - ☐ CPU Context switching
 - ☐ Week 4: Lecture notes
 - ☐ Week 4 Feedback Form : Introduction to Operating Systems!
 - ☒ Quiz: Week 4 : Assignment 4
- Week 5: Scheduling
 - ☐ CPU Scheduling
 - ☐ Priority Based Scheduling Algorithms
 - ☐ Multi-Processor Scheduling

Assessment ChatBot
Bookmark Full Store

Week 4 : Assignment 4

The due date for submitting this assignment has passed.Your last recorded submission was on 2025-08-11, 18:59 IST.

Due date on 2026-08-19, 23:59

1. A context switch occurs when a state of process changes from new to ready
1 Point

☐ True
 ☒ False

Yes, the answer is correct. Score: 1
Accepted answer is False

2. A thermal sensor in the CPU checks the temperature of the machine, if it is more than 40°C an interrupt is generated to shutdown the machine. This interrupt is a hardware interrupt.
1 Point

☒ True
 ☐ False

Yes, the answer is correct. Score: 1
Accepted answer is True

3. What is the correct order of the following events
1 Point

I. device raises interrupt
 II. CPU checks signal on INT pin and sends INT ACK signal
 III. Interrupt Handler Routine is executed
 IV. Interrupt Controller sends INT signal to CPU
 V. CPU identifies which device has raised the interrupt

☐ I,II,III,IV,V
 ☒ V,I,V,II,III

swayamNPTEL

Powered WidgetsAbout the courseAnnouncementsMy BookmarksQ&AManage Exam

Introduction To Operating Systems

Course Progress50.55% (3 of 7)

Introduction

Create Execute and Exit from Processes

System Calls for Process Management

Week 3: Lecture notes

Week 3 Feedback Form : Introduction to Operating Systems!

Quiz: Week 3 : Assignment 3

Week 3: Assignment Solution

Week 4: Interrupts

Interrupts

Interrupt Handling

Software Interrupts and System calls

CPU Context switching

Week 4: Lecture notes

Week 4 Feedback Form : Introduction to Operating Systems!

Quiz: Week 4 : Assignment 4

Week 5: Scheduling

CPU Scheduling

Priority Based Scheduling Algorithms

Multi-Processor Scheduling

Introduction To Operating Systems

Course Progress50.55% (3 of 7)

Introduction

Create Execute and Exit from Processes

System Calls for Process Management

Week 3: Lecture notes

Week 3 Feedback Form : Introduction to Operating Systems!

Quiz: Week 3 : Assignment 3

Week 3: Assignment Solution

Week 4: Interrupts

Interrupts

Interrupt Handling

Software Interrupts and System calls

CPU Context switching

Week 4: Lecture notes

Week 4 Feedback Form : Introduction to Operating Systems!

Quiz: Week 4 : Assignment 4

Week 5: Scheduling

CPU Scheduling

Priority Based Scheduling Algorithms

AssessmentChatBot

Week 4 : Assignment 4

The due date for submitting this assignment has passed. Your last recorded submission was on 2025-08-11, 18:59 IST.

BookmarkFull Screen

Due date on 2025-08-19, 23:59 IST

Yes, the answer is correct. Score: 1

Accepted answer is False

5. Match the following Mnemonics of Interrupt Vector used in x86 to the type of interrupt the are

1 Point

I. #DF a. Trap
II. #DP b. Fault
III. #SS c. Abort

☐ I - a II - b III - c

☒ I - a II - c III - b

☐ I - b II - c III - a

☐ I - c II - b III - a

Yes, the answer is correct. Score: 1

Accepted answer is I - a II - c III - b

6. A CPU with four processes executing has an Operating System with a non-preemptive scheduler. What is the total number of scheduling possible for the 4 processes? _____

1 Point

24

Yes, the answer is correct. Score: 1

Accepted answer is Type: Numeric: 24

7. More than one System call in an operating system can have the same system call number

1 Point

AssessmentChatBot

Week 4 : Assignment 4

The due date for submitting this assignment has passed. Your last recorded submission was on 2025-08-11, 18:59 IST.

BookmarkFull Screen

Due date on 2025-08-19, 21

☒ False

Yes, the answer is correct. Score: 1

Accepted answer is False

9. Read the following statements and point out which are true and false

1 Point

I. A context Switch can occur when a process moves from a blocked to ready state
II. A context Switch can occur when a process moves from a running to zombie state
III. A context Switch can occur when a process moves from a running to ready state

☒ I - False , II - True, III - True

☐ I - False , II - False, III - True

☐ I - True, II - True, III - True

☐ I - False, II - True, III - False

Yes, the answer is correct. Score: 1

Accepted answer is I - False , II - True, III - True

10. Timer Interrupt Latency should be as high as possible for smooth context switching

1 Point

☐ True

☒ False

Yes, the answer is correct. Score: 1

Accepted answer is False